Proposed Solution

This document describes OptiCrop's proposed solution in detail, covering the product concept, feature set, user experience design, and technical approach.

Solution Overview

OptiCrop is a machine learning-powered crop recommendation web application. A farmer enters seven soil and climate parameters; OptiCrop's Random Forest classifier analyses these against patterns learned from 2,200 labelled agricultural samples and returns the optimal crop to plant, along with a confidence score and growing tips.

ML Training Pipeline

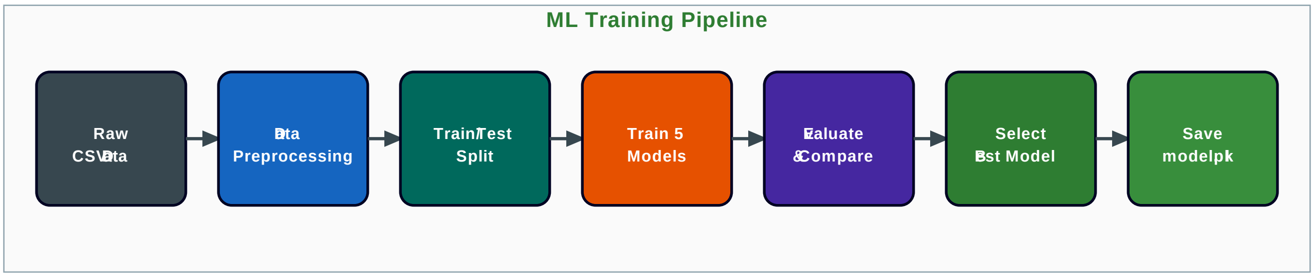

Figure 2: End-to-End ML Training Pipeline

Core Feature Set

Feature	Description	User Benefit
Smart Crop Predictor	7-input ML form instant crop recommendation	Decides in seconds what took seasons to learn
Confidence Score	Probability (%) of the top recommendation	Farmer can judge how certain the advice is
Top-3 Crop List	Second and third best crop alternatives displayed	Fallback options if primary crop seeds unavailable
Crop Profile Card	Image, description, growing season, water needs	One-stop info no need to search elsewhere
Input Tooltip Guide	Contextual help for each of the 7 parameters	Reduces data entry errors and user confusion
Prediction Log	CSV file of all predictions with timestamp	Researcher / admin audit trail
Mobile-Responsive UI	Bootstrap 5 layout adapts to any screen size	Accessible on ▲6,000 budget Android phones

ML Model Comparison

Algorithm	Accuracy	Precision	Recall	F1-Score	Selected
K-Nearest Neighbours (K=5)	97.5%	97.4%	97.5%	97.4%	No
Logistic Regression	95.2%	95.1%	95.2%	95.1%	No
Decision Tree	98.6%	98.6%	98.6%	98.6%	No
Random Forest (n=100)	99.09%	99.1%	99.1%	99.1%	YES
K-Means Clustering					No (unsupervised)

Supported Crops (22 Classes)

rice, maize, chickpea, kidneybeans, pigeonpeas, mothbeans, mungbean, blackgram, lentil, pomegranate, banana, mango, grapes, watermelon, muskmelon, apple, orange, papaya, coconut, cotton, jute, coffee